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Deep eutectic solvents-based adsorbents in environmental analysis

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ABSTRACT

Deep eutectic solvents (DESs), a new class of environmentally friendly solvents, are gaining more attention from researchers for extracting and separating targeted analytes from different complex samples. DESs are liquid at room temperature and contain a hydrogen bond donor and a hydrogen bond acceptor. DESs are non-volatile solvents, have excellent thermal and electrical stability, and can easily dissolve a number of organic and inorganic substances. DESs have attracted a lot of attention as new green solvents are expected to take the place of traditional organic solvents. DESs are sustainable solvents that concern significant interest from researchers and various industries. They are non-flammable, chemically stable, biocompatible, biodegradable, and have low volatile content. DESs have tunable physicochemical properties that can be tailored to meet the needs of a definite task. This review summarizes the DESs-based adsorbent materials and its preparation characterizations and application by highlighting the unique properties such as extraction and separation applications. To summarize the DESs-based applications, this review reported up-to-date data related to this field. and also provide an easy and single path to cover the DESs-based materials and its adsorption applications. This review exploits new DESs and DESs-based materials that are diversifing into extraction and separation due to the development of DESs.

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1. Introduction

Development in deep eutectic solvents (DESs) represents a significant advancement in green chemistry. DESs are typically characterized as binary or ternary combinations of substances that can associate primarily through hydrogen bonds. DESs are widely used around the world due to their unique characteristics. The term DESs was coined by Abbott et al., in 2003 [1]. Due to their importance and exciting potential, these solvents have emerged as a dynamic, novel, and interesting “green chemistry” subfield. DESs are an advanced class of ionic liquids (ILs); their synthesis is as simple as just combining two or more substances made up of hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs). HBAs are quartet phosphonium, ammonium, and mineral salts. HBDs are alcohols, organic acids, amides, and sucrose. DESs are molten in nature and prepared from a eutectic mixture of two or more harmless constituents that play a significant role as HBAs or HBDs-based new generation solvents. In general, DES contains large asymmetrical ions with lower lattice energy. The decrease in the mixture's melting point relative to each component's melting point is caused by the charge delocalization of the hydrogen bond between the hydrogen bond donor moiety and halide ion. DESs have some distinguishing characteristics, including simple and green synthetic processes, cost-effectiveness, environmentally friendly, with high efficiency, making them suitable for large-scale production. According to Abbott et al., the scheme and preparation of DESs are still in their early stages, and more research is required. DESs is a hot research era in different workplaces with novel and emerging applications, especially in chemistry, metals, biomaterials, separation sciences, electrochemistry, organocatalysis, solar technology, and photosynthesis. DESs have tunable properties with excellent polarities due to their arrangement's possible hydrophobic or hydrophilic constituents. In addition, DESs prepared from renewable sources have low toxicity with higher biodegradability. The ammonium-based quaternary salt has many advantages during the preparation of DESs, including low raw material cost,

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chemical stability with water, simple synthesis process, biodegradability, biocompatible, and low-toxic or non-toxicity. The benefits listed above all contribute to DESs' environmental friendliness. Since realizing the potential for viable chemical methodologies, in the last decade, researchers have focused on DESs [2]. Some particular DESs types have also received much attention, such as natural-DESs (NADESs) were proposed by Choi et al., in 2011. The eutectic mixtures are designed by principal cellular components, including alcohols, sugar, carboxylic acid, and amino acids. They hypothesized that the NADESs might play an essential role in a wide range of cellular processes that explain the mechanism and phenomena that would otherwise be difficult to comprehend, like the bio-synthesis of nonpolar smaller and larger molecules. This recommends that lipids and water not only are solvents in biospheres but also explain the bioactive compounds with high solubilizing capacity. Because of the natural component, NADESs have a significant advantage over DESs because they fully represent green chemistry principles. Surprisingly, in recent years, there has been an exponential growth of publications focusing on using environmentally friendly solvents such as DESs and water, both of which follow the green chemistry principles. DESs have unique physicochemical characteristics (e.g., thermal stability, negligible vapor pressure, low cost, non-flammable, and a stronger hydrogen bond network. Intriguingly, DESs have another feature: they play a vital role in nature's game. NADESs are prepared from the primary metabolites (e.g., amino acids, sugars, carboxylic acids, vitamins, and natural amino alcohols). NADESs may allow biological processes to occur in living organisms in the interim as the liquid state in addition to lipid and water, assisting the storage and conveyance of poorly water-soluble macro-molecules and protecting biological systems from cold and drought stresses. Simultaneously, water plays an essential role in preparing NADESs by altering the physicochemical characteristics of the corresponding supramolecular network. As a result, NADESs interactions with natural and biomaterials are altered (vide infra). Although a bit of water in hydrated DESs would strengthen the hydrogen-bond network, researchers have reported that the DESs nanostructure retains up to 42 wt percent of water. At 51% water weight, the solution begins to behave almost like solutes in an aqueous solution. Despite the similarities between DESs and water, when tested individually in specific reactions, these “solvents” in nature sometimes offer complimentary performances on organic reactivity. The reasons are still unknown, and researchers have considered understanding molecular mechanisms of action a hotly debated topic. The aim is to provide the readers with an overview of the wide range of possible reactions and the potential benefits of using DESs in (bio) synthetic processes synergistically or singly. The interested readers are referred to the comprehensive reviews/monographies already available in the literature. This review assists the scientific community in studying the DESs-based adsorbent material to eliminate toxic pollutants from the environmental samples. This review focus on the new development in the DESs-based adsorbent material, especially in the synthesis, characterization, and environmental applications. It benefits multidisciplinary researchers and offers new ideas in different fields, including analytical chemistry, environmental, pharmaceutical, and biological.

2. Chemistry of DESs

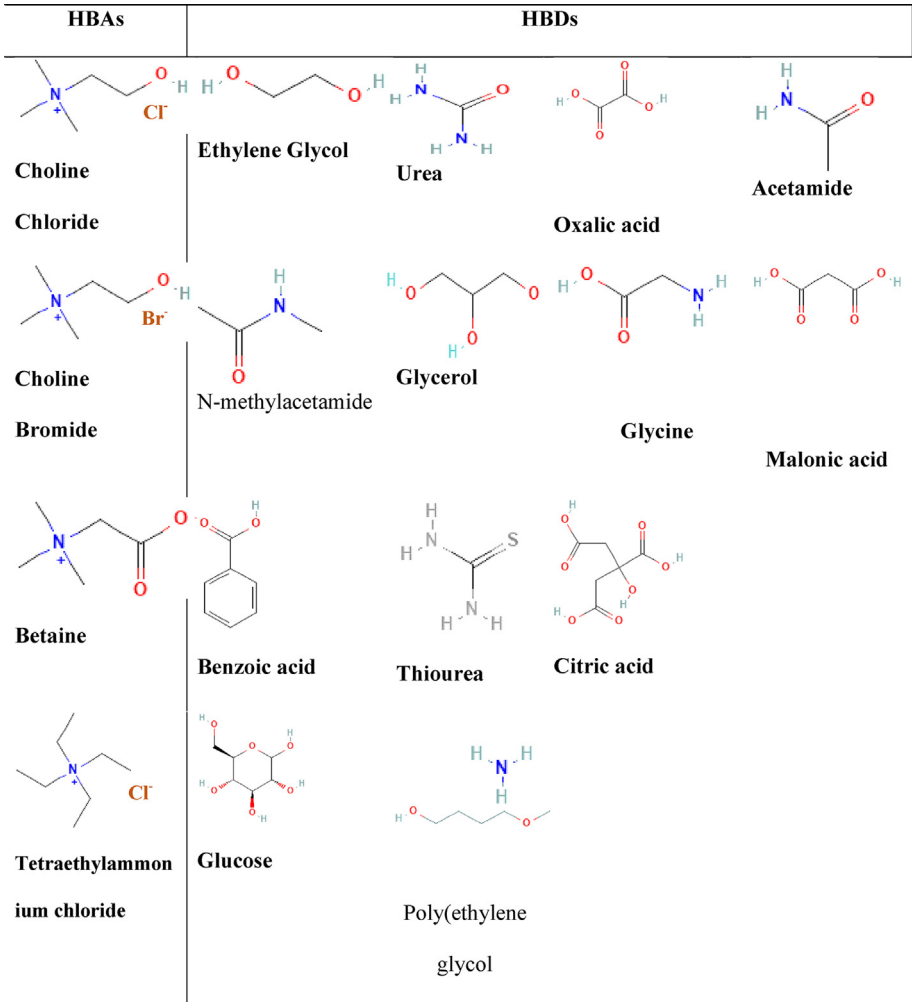
DESs are unique solvents with unusual characteristics; the terms have been used interchangeably in many publications. DESs are fabricated as a eutectic mixture by combining Lewis or Bronsted acid and a base containing a mixture of anions and cations species. Furthermore, a eutectic system is a chemical compound mixture that exhibits a single chemical composition that freezes at lower temperatures. The eutectic fact is given by joining the eutectic composition and temperature; due to different solvent properties and the fact that some DES constituents are not ionic, the term DESs was coined to distinguish them from ILs. DESs drop the freezing point in the eutectic mixture, which can be as much as a few Celsius. To synthesize a eutectic mixture, two or more constituents are mixed at a specific temperature. HBAs and HBDs can form a complex with the combined quaternary ammonium salt halide ion to form DESs. The interaction between HBAs and HBDs components has the potential to link them together via hydrogen bond interactions. DESs are classified based on their physicochemical properties, particularly their melting point, which should be lower than the melting point of each distinct component. DESs typically have a large freezing point depression and are liquid at temperatures below 423 K, most of which are liquid between room temperature and 343 K. In 2007, Abbott defined DES as $R_1R_2R_3$- R_4N^+ X^-:Y^-, where X is a halide ion (often Cl^-), and Y is a complexing agent composed of Cat^+X^- [3]. According to the literature, interest in DESs is significantly growing, particularly in various chemical, electrochemical and separation applications. Different types of salts and HBDs, or complexing agents, have been reported for these applications. Because of its low cost, eco-friendliness, and non-hazardous properties, ChCl has been widely used as an organic salt in preparation of eutectic solvents, typically in conjunction with economical and non-toxic HBDs such as urea, glycerol, ethylene glycol, or carboxylic acids. Table 1 shows the various types of HBA and HBD commonly used to form DESs. These DESs combinations are appealing because they have comparable physicochemical properties to traditional imidazolium-based ILs and can thus replace them in many applications. Compared to conventional organic solvents, DESs are known as environmentally friendly and non-volatile organic solvents and non-flammable, making storage convenient.

3. Green chemistry scenario

With the advent of green chemistry, using eco-friendly and non-hazardous solvents has become mandatory. The concept of “green” solvents indicates the desire to reduce the environmental effect caused by using solvents in manufacturing chemicals. Several solvents that might be utilized in the extraction process may be referred to as “green solvents” in light of this. Green solvents are non-volatile, non-toxic, recyclable, biodegradable, and cost-effective. Fig. 1 shows different solvents that meet this requirement. They are classified into four different types such as supercritical fluids, bio-based, supramolecular, and neoteric solvents. It is not simple to switch out a hazardous solvent for a more environmentally friendly substitute in a separation process; in some instances, new challenges and limitations can appear because of the different physicochemical characteristics of the solvents under consideration. Supercritical fluids (SCFs) are substances that have pressure and temperature values that are higher than their critical values. Gas-liquid characteristics, such as liquid density, solvating properties, gas-like viscosity, and diffusivity, define SCFs. As a result, they are ideal solvents for supercritical fluid extraction (SFC) operations. Solvents from renewable biomass sources, such as plant products, energy crops, waste materials, and aquatic biomass, are “bio-based solvents” prepared in the biorefinery, which seeks maximal extraction recovery and the creation of high-value products. Nanostructured liquids are known as supramolecular solvents (SUPRASs). SUPRASs are fabricated by the colloidal suspensions of amphiphiles through the spontaneous sequential mechanisms of self-assembly and coacervation. Fig. 1 shows the different types of green solvents.

The definition of coacervation is “the separation of colloidal

Table 1
Commonly used different HBA and HBD to form DESs.



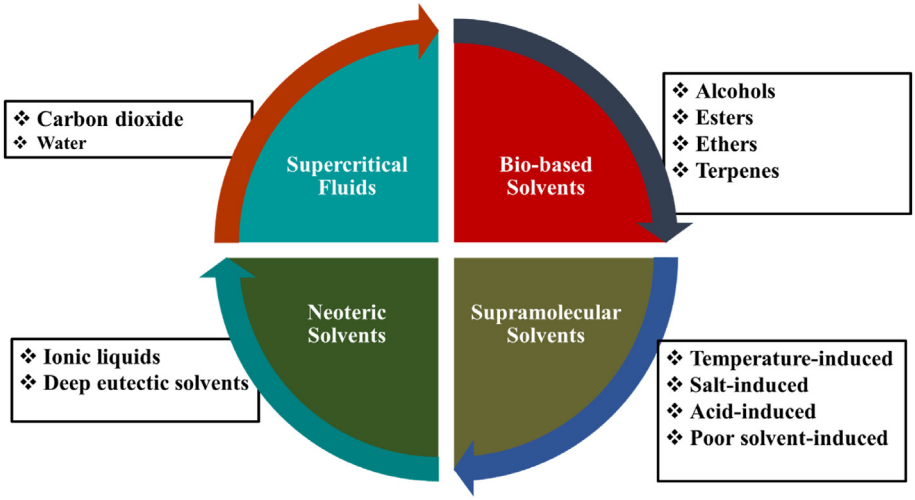


Fig. 1. Different types of green solvents.

systems into two liquid phases." Neoteric solvents have structurally unique or unusual physical and chemical properties that can be tailored for various applications by altering the chemical ingredients [50]. Fluorous solvents, ionic liquids, and eutectic solvents have attracted the most attention among neoteric solvents. DESs were developed to address the environmental problems associated with ILs. They are similar to ionic liquids in their physical and chemical properties but are easier to make, more stable, and cheaper. Most of them are also good for the environment. In terms of new green extraction technologies, DESs have demonstrated significant promise, and in the years to come, it is anticipated that the industry will widely adopt them.

DESs, an alternative to ILs, have similar properties to ILs but cost less to make since the raw ingredients are less expensive, less hazardous, and frequently biodegradable. For several decades, DESs, particularly those comprised HBDs and HBAs species, have been regarded as green and environmentally friendly for certain characteristics. First, the intercalation of HBDs and HBAs species can generate liquids that are non-volatile, non-hazardous to the environment, and practical. In addition, the III and advanced types of DESs consist of metal chlorides and NADESs consisting of tertiary amines, carboxylic acids, and alcohols, which are frequently biodegradable and non-toxic, and their low reactivity with water makes them ideal solvents for a variety of analytical applications. In addition, their low production costs and wide availability of raw materials. Most DESs are solid at room temperature and have a high viscosity, which limits their use as extraction solvents. NADESs, a new class of extraction solvents, have just been introduced to the analytical practice. In addition to all of the benefits of DES, NADESs are even more environmentally favorable because they are derived from natural sources. The most promising and popular green extraction solvents DESs are reviewed in this research [4].

4. Classification of DESs

DESs were divided into four categories based on the general formula Cat + X zY, where Cat + indicates ammonium, phosphonium, or sulfonium and X is a Lewis base, typically a halide anion, in order to distinguish between the potential eutectics. The amount of Y molecules interacting with the associated anion is denoted by the symbol Y, which stands for a Lewis or Brnsted acid. Types of DESs such as I, II, III and IV. DESs comprise metal salts (hydrated metal chlorides) and other neutral compounds or organic salts. The most extensively researched eutectics are type III, which typically use choline chloride and a variety of hydrogen bond donors. Choline chloride has been widely used because it is reasonably inexpensive, non-toxic, and biodegradable. In actuality, choline chloride was the main component of the first type III deep eutectic solvent. Quaternary ammonium or phosphonium salts are the common hydrogen bond donors, while amides, alcohols, and carboxylic acids are the most typical hydrogen bond, acceptors. For manufacturing NADESs, substances including sugars, sugar alcohols, and amino acids are also considered. Hydrophobic DESs were introduced more recently. They work by using long alkyl chain alcohols and carboxylic acids as hydrogen bond donors, along with bromide, menthol, thymol, and fatty acids as hydrogen bond acceptors. Fig. 2 shows the classification of DESs. Additionally, active pharmacological components, including ibuprofen, lidocaine, and phenylacetic acid, can be used to prepare DESs. The solvents of hydrophobic substances, such as tetrabutylammonium, are then referred to as therapeutic DESs. On the other hand, it is occasionally possible to classify NADESs as type III deep eutectic solvents. A special type of DESs called NADESs was proposed during the development process described above. NADESs are made up of CHCl and natural components, such as water. In the NADESs system, water helps to make the oil less thick. Choi et al. said the NADES system was made through the hydrogen bonding between the different parts. Most of the time, the HBAs comprise amines or amino acids, and the HBDs contain carbohydrates. So, the NADES are environmentally friendly and cheaper than other type III DESs. DESs are water-soluble because almost all of their parts have several hydroxyls, carboxyl, or amino groups [5].

5. Preparation of DESs

DESs are synthesized using natural and renewable constituents and have analogous physicochemical characteristics to ILs. DESs are typically made up of bi- or tri-components of hydrogen bond donor and acceptor, such as acid amides and quaternary ammonium salts with lower melting points [6]. The easiest way to prepare DESs, which use all their atoms, is to mix HBAs and HBDs. The most common way to make DESs called the "heating and stirring method," is to mix and stir two or more components at a higher temperature (ordinarily 80 °C) until homogeneous liquid forms. This method was suggested by Abbott et al., in 2003. Different research groups use heating and stirring methods, and the reaction temperature (50–100 °C) is usually changed to fit the other reaction parts. Most DESs contain quaternary ammonium salt, which means that the DESs made by heating and stirring will have some water. Gutierrez et al. [7] came up with two other ways to make DESs using evaporating and freeze-drying processes) that don't involve water. Florindo et al. [8] also developed a way to make DESs that didn't affect high temperatures. When DESs (choline chloride: glutaric) is made by grinding, they are purer than when made by heating and stirring. The reaction between the carboxylic acid and the component with an OH group could also lead to the formation of esters.

6. Characterization

DESs characterization is also an essential part of a study to check the formation of newly prepared DESs different techniques have been used, such as Proton nuclear magnetic resonance (1H NMR) and Fourier transform infrared (FTIR). 1H NMR spectroscopy can investigate hydrogen-bonding interactions and the formation of DES-specific halide ion hydrogen bond-donor supramolecular complexes. FTIR spectroscopy is a useful tool for determining hydrogen bonding and active functionality in DESs; a number of changes happened. In order to learn more information about the potential interactions, structural changes, and hydrogen bond formation in DESs, it is helpful to compare the FTIR spectrum of DESs with each of its constituent parts. The DESs' FTIR spectra showed several differences.

7. Physical and chemical properties of DESs

The physical and chemical characteristics of DESs also influence the adsorption process. During the elimination procedure, volatile solvents are typically the most common media utilized, particularly for liquid and solid-liquid extraction. On the other hand, certain compounds degrade when exposed to volatile solvents, making the elimination and adsorption of target analytes very challenging. In addition, their quantitative analysis will invariably result in unforeseen errors. Methanol, dichloromethane, toluene, and acetonitrile are organic solvents with higher toxicity levels. These solvents have different affinities to water than other organic solvents. On the other hand, several extraction techniques have limitations, such as additional energy sources, heat, microwaves, or ultrasound, ultimately leading to the evaporation of organic solvents. As a result, to successfully replace conventional organic

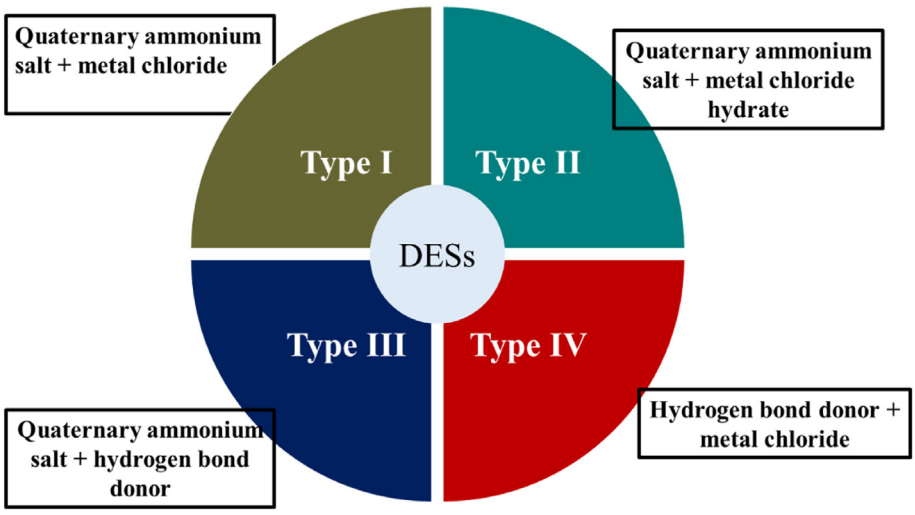


Fig. 2. Classification of DESs.

solvents, it is essential to develop extraction media that are both low in toxicity and vapor pressure and have a similar physicochemical performance in the solvent system. Due to these limitations, the development of DESs has garnered significant attention. The most important factors that influence the physicochemical properties of DESs are the combination structure of HBDs and HBAs. During the extraction process, different arrangements of HBDs and HBAs lead to different intermolecular forces (i.e., H-bonding, p-p interaction, and ionic bonding) between the targets and DESs. The density, viscosity, and melting point of hydrophilic and hydrophobic are the physicochemical properties of DESs [9].

The interest in studying the physical and chemical characteristics of DESs has also developed rapidly. However, few existing studies on physicochemical properties are reported in the literature. As a result, the following sections will study the available literature on the physicochemical characteristics of DESs and their applications. The properties of the DESs greatly influence the extraction efficiency. These properties are determined by the physicochemical properties of each DESs constituent and the interactions between the two components. The elimination efficiency is also influenced by the DESs solubilizing capacities and, eventually, the ability to disrupt cell structures. Water content is typically influenced during the extraction of targeted bioactive compounds and also influences Elimination efficiency. The water contents can significantly affect the properties of DESs [10]. DESs are solvents that can be chemically tailored due to their ability to be synthesized by combining various quaternary ammonium salts with several HBDs. Different cations and anions can be used to make different DESs, allowing for greater control over their water compatibility. They are denser than water, miscible with high-polarity substances, and can dissolve inorganic and organic constituents. DESs have many advantageous properties in chemistry and physics, including moderate viscosity, low freezing point, density, conductivity, and high thermal stability. The main physicochemical properties of DESs will be described and discussed in this section.

7.1. Melting point (MP)

DESs, as previously stated, have a lower melting point than any of their compounds. However, DESs are liquid at 25 °C; thus, the melting temperature should be the first factor to consider when selecting DESs for an extraction method. The inter-molecular forces (hydrogen bonds and van der Waals force) are present among the HBA and HBD that affect the MP of DESs. Optimizing the molar ratio of HBA to HBD throughout the experiment to investigate the working mechanism together. ChCl is often used as the HBAs for hydrophilic DESs that are used often. Abbott et al. [1] found that a 1:2 mole ratio of ChCl and urea could combine to make DESs with a low MP up to 12 °C, which shows a relationship with mole ratios. Abbott et al. also made a chain of DESs out of ChCl and HBDs (malonic acid, oxalic acid, and succinic acid) at different eutectic temperatures, such as 10 °C 34 °C, and 71 °C, respectively. This shows that HBD plays a key role in the melting point of DESs. As far as we know, the significant drop in melting point is caused by the binding between the anion of the HBA (i.e., methyl triphenylphosphonium bromide, ChCl, benzyltriphenylphosphonium chloride, tetrabutylammonium bromide and N, *N*-diethylethanolammonium chloride) HBDs are produced by changes in entropy and lattice energies. As estimated, the MP of highly reported hydrophobic DESs was below 20 °C. This means they can stay in the liquid phase at room temperature and be used as extraction media. Osch et al. [11] proposed that the MP of DESs heavily depends on the alkyl chain structure of HBDs and HBAs. The melting points of both ionic and non-ionic hydrophobic DESs go up as the alkyl chain length in the fatty acid HBDs increases. Also, anions and cations in the ionic components HBAs of ionic, hydrophobic DESs greatly affect their melting point. The melting point increases as the length of the alkyl chain in the ammonium cation increases: N4444eCl > N8881eCl > N8888eCl. The melting point of hydrophobic DESs is affected by the electronegativity of the anion in HBAs (Cl or Br): N4444eCl (11.95C) N4444eBr (16C), N8881eCl (0.05C) N8881eBr (8.95C), N8888eCl (1.95C) N8888eBr (8.95C). On the other hand, it was hard to determine the PM of non-ionic hydrophobic DESs by looking at their full solid-liquid behavior. Martins et al. devised a new way to measure the melting point using phase diagrams. Differential scanning calorimetry measured the solid-liquid equilibria phase diagrams of mixtures comprising menthol, thymol and monocarboxylic acids and terpenes over the entire structure range, offering a wide configuration in the liquid phase than before reported [12].

7.2. Viscosity

The viscosity of DESs is a crucial factor in the adsorption procedure, especially for extensive applications. The HBDs parts of the hydrophilic DESs system were responsible for most of the changes in

viscosity. DESs that are hydrophilic and the phenol, glycols and ethylene glycol are less viscous. On the other hand, DESs made of sugar, polycarboxylic acids, or urea-based on ChCl have high viscosity or medium. The high viscosity of DESs is caused by hydrogen bonds between the HBA and HBD, making it hard for DESs molecules to move around. The ones with the fewest ions were the ionic, hydrophobic DESs with the highest viscosity (>172.87 mPa$s). On the other side, non-ionic DESs made with OctA and DoDecA were said to have a low viscosity of 7.085 mPa$s. A number of non-ionic hydrophobic DESs have low viscosity (218.927 mPa$s) than ionic, hydrophobic DESs. In general, the chemical structure of HBAs affects the viscosity of most DESs in the following order: long-chain fatty acid, thymol, menthol, and quaternary ammonium salts. Because of coulombic forces, quaternary ammonium salt DESs have a higher viscosity. Also, the high viscosity of hydrophilic DESs isn't a problem because the non-ionic hydrophilic DESs-based on thymol, and menthol, DoDecA, have a very low viscosity (7.085e13.2 mPa$s). No matter how water-friendly or water-repellent DESs are, their viscosity is still not as good as organic solvents like heptane z 0.4 mPa$s [13].

7.3. Density

Density is also a fundamental physical property of DESs; density plays an important role in solvent design and selection in the extraction process. In general, the density of DESs depends on how their molecules are packed together and how strongly their parts interact. Most DESs have lower densities than water, and some have the same densities as water, which is better for the sample pretreatment and preparation process. Ibrahim et al. [14] said that the density of DESs changes with temperature. At room temperature, MTPB/EG had a density of 1.25 g/cm^3, but at 80 °C, it had a density of 1.21 g/cm^3. At 80 °C, the densities of the DESs of N4444eBr/EG and ChCl/EG were 1.03 g/cm^3 and 1.09 g/cm^3, respectively. Martins et al. found that as the length of the monocarboxylic acid's alkyl chain increases, the density of DESs made of menthol and monocarboxylic acid decreases. In contrast to DESs made from menthol, DESs made from thymol get denser as the length of the alkyl chain in the monocarboxylic acid increases.

7.4. Surface tension

DESs' surface tension is a crucial characteristic since it is closely related to the strength of the intermolecular interactions between the hydrogen bond donor and the associated salt. High surface tension is seen in highly viscous liquids, though. The values concerning the reported DESs typically range between 35 and 75 mN m1 at 25 °C. For sugar-based DESs such as choline chloride: D-glucose and choline chloride: D-fructose, again, exceptional high values were noted, reflecting their vast hydrogen-bond network. Additionally, the surface tension is influenced by the cation type and salt mole fraction since quaternary ammonium salts with more hydroxyl groups or longer alkyl chains have greater surface tensions. As the temperature rises, the surface tension also linearly decreases. Surface tension is affected by the strength of intermolecular interactions in DESs and temperature. The surface tension of most DESs is higher than that of conventional solvents. In extraction techniques, DESs with a low surface tension could be applied as a wetting or adhesive agent.

7.5. Polarity and solubilizing properties

The polarity of a DESs is essential for its miscibility in the sample solution or other solvents and its solubilizing proficiency toward the targeted analytes. DESs are well known for dissolving polar and weak-polar compounds such as drugs, pharmaceutical ingredients, metal oxides, carbon dioxide, and elemental species such as lead, mercury, and cadmium. Furthermore, DESs demonstrated a better solubilizing effect and thus better extractability when compared to conventional solvents. These characteristics can be reorganized moderately by the “like dissolves like” theory, which aids in selecting the appropriate DESs in terms of polarity. DESs have recently been tailored to specific by selecting distinct components based on the analytes using silico procedures. The novel interactions of DESs with target analytes allow for the selective separation of traces of this analyte from complex matrices. It should be noted that increasing the temperature generally decreases the polarity of DESs due to a decrease in the solvent's hydrogen-bond donating acidity, as observed with choline chloride/glycerol DESs [15].

7.6. pH

The pH of the DESs can affect the charge of the bioactive compounds during the elimination process and thus the adsorption efficiency. Analytes are generally much easier to extract in their neutral forms using weakly polar solvents. As a result, the extraction procedure pH should be higher than or close to the pKa values of the analytes being studied. The pH of various DESs varies with temperature, and the type of hydrogen bond donor heavily influences their acidity. Because highly polar molecules require an acidic environment for extraction, organic acid- DESs or some natural DESs are frequently the best choices. This was seen with organic acid-based natural DESs, which produced the best extraction results for anthocyanin (polar compounds). In contrast, sugar-based natural DESs were better for other phenolic compounds. pH highly influences protein extraction because of its amphoteric properties, and their isoelectric point should be considered [16].

7.7. Hydrogen bond

Hydrogen bonding is a unique form of dipole-dipole attraction between molecules. It results from the attractive force between hydrogen atoms covalently bound to a highly electronegative atom. Hydrogen bond plays a crucial role in the formation and stability of DESs. DESs have been produced utilizing a variety of hydrogen bond donors, including carboxylic acids, alcohols, and amides. These liquids are easy to make and relatively unreactive with water; many are biodegradable and inexpensive. DESs are exceptionally flexible because of the wide range of hydrogen bond donors available. The hydrogen bond donor determines the solvent's physical properties and can be easily modified for specific uses. DESs have been remarkably adaptable, with a wide range of potential applications studied, including the removal of glycerol from biodiesel metal oxide processing and cellulose derivative synthesis. DESs have lower melting pints than the melting points of the separate components because the charge delocalization induced by hydrogen bonding between hydrogen-donor moiety and halide ion is responsible for the lower melting point of the solvent. Because of their capacity to solvate a wide range of transition metal species, including chlorides and oxides, type III eutectics produced from choline chloride and hydrogen bond donors have attracted the researchers' intention. When these HBDs are multifunctional, the eutectic point tends toward a 1:1 M ratio of salt to HBD. The same study determined that the freezing point depression is proportional to the mass fraction of HBD in the mixture. Using the atomic contribution method, the refractive index of choline chloride-based liquids with the hydrogen bond donors glycerol, ethylene glycol, and 1,4-butanediol was measured and compared to the predicted values. Using the chosen method, it was discovered that it was

possible to predict the refractive indices of these solvents and that the refractive index lay between the values for salt and HBD. Considering the relationship between viscosity and temperature, it is not surprising that the electrical conductivity of DESs varies with temperature. The effective polarity of choline chloride combined with four HBDs, glycerol, 1,2-ethanediol, urea, and malonic acid, has been predicted, and it was discovered that all four liquids are relatively dipolar. After further investigation, the authors hypothesized that this was due to the formation of a film on the Hg electrode's surface. It was discovered that the negative limit of polarisation depends on the hydrogen bond donor, with the limit increases in the following order: 1,3-propanediol < urea < ethylene glycol <1,2-propanediol. Increasing DES temperature increased the capacitive current observed and diminished the electrochemical window.

8. Application of DESs in environmental analysis

The advent of DESs to broaden their role as an extractant is a critical breakthrough in developing sustainable extraction initiatives. DESs have distinguished themselves by virtue of their biodegradability, low cost, and ease of synthesis. Currently, DESs are hydrophilic, limiting the range of target compounds that DESs can extract targeted analytes. DESs are a new class of solvents due to their benefits in terms of safety, simplicity, sustainability, and cost, as well as a wide range of applications such as preparation and functionalization of nanomaterials, soluble drugs, biodiesel purification of glycerol, organic synthesis, electrodeposition, and metal processing [17]. They, are being used as a “greener alternative” to traditional solvents because they are biodegradable, non-toxic, and safe. DESs are generally thought to be environmentally friendly; however, only one of their four types (type III) can be considered truly “green” due to the absence of metal salts in their composition. Many DESs demonstrated high compatibility with various instrumental detection systems, including UV−visible spectroscopy, mass spectrometry, flame ionization detector, fluorescence spectroscopy diode-array-detector, Fourier-transform infrared spectroscopy, and graphite furnace atomic absorption spectroscopy, As a result, hyphenated with advanced separation techniques such as gas chromatography, high-performance liquid chromatography, and thin-layer chromatography are being developed, using a certain percentage of DESs in the mobile phase. DESs, which has almost no toxicity, high recyclability, low in-flammability, and volatility, and is completely biodegradable, are promising sustainable alternative to ionic liquids and conventional solvents [18]. DESs are not only environmentally friendly, but they are also simple to prepare. These fascinating physical properties and the freedom to design DESs play an essential role in various fields, such as separation, sample preparation, extraction, biotransformation, nanomaterial, electrochemistry, and interactions with other compounds via surface modification, providing a suitable platform for functionalization, synthesis, and development of adsorbents with green chemistry framework. Functionalization of carbon nanotubes, graphene and graphene oxide, carbon monoliths and synthesis of composite materials have potential applications of DESs in water treatment [19,20]. Choline Chloride-Glycerol (ChCl-Gly) is a common type III DES. It was also discovered that aqueous ChCl-Gly is one of the most easily biodegradable DESs. The biodegradability of ChCl-Gly after one week and 28 days was 84 and 91%, respectively. As a result, among the DESs, ChCl-Gly could be regarded as a “green solvent.” Carbon nanotubes (CNTs) have long been recognized as promising for CO_2 adsorption due to their high surface area, tunable pore structure, high amenability to surface modification, and reusability. Despite having the same surface area, some studies show that CNTs have a higher adsorption capacity than silica materials and activated carbon. DESs have recently received much attention because of their promising applications in metal processing, green adsorption technology, nanomaterials, and catalysis [21]. A recent study found that the choline chloride-oxalic acid DESs were selective for copper extraction in sediment samples. Another study described the microextraction of tartrazine, an artificial dye, at trace levels using tetrabutylammonium chloride/bromide-decanoic acid DES. DESs are regarded as an appealing alternative because it retains the benefits of ILs in terms of tunability while overcoming the shortcomings of ionic liquids in terms of sustainability as a green solvent and operational simplicity. It was demonstrated that the extraction yield of polysaccharides obtained using the optimal DESs (consisting of 70% water plus choline chloride/glycerol in a 1:2 M ratio) was at least twice as high as the yield obtained from a solution of HCl/water mixture, which is the standard method for extracting polysaccharides. Machmudah et al. [22] demonstrated the resulting subcritical water extraction method's efficacy in extracting xanthones by adding 10−30% citric acid/alanine (1:1 M ratio) DESs to water media. As a result, scanning electron microscope images revealed surface disruption of mangosteen pericarps after treatment with this method at high temperatures. DESs cleaved intermolecular and intramolecular bonds in and between lignin, cellulose, and hemicellulose, forming pores. Recent research examined the effectiveness of fatty acids in removing a variety of micropollutants. Fatty acid-based DESs, in particular, have been used to successfully extract Bisphenol A and pesticides with recovery efficiencies of up to 92% and 80%, respectively. Despite the benefits of fatty acid-based DESs, few experiments have been conducted, and the underlying chemistry behind their performance is still unknown. So far, research on the potential of DESs based solely on fatty acids as efficient dye extractants Rhodamine B, Eriochrome Black T, and Patent Blue V, as well as a family of food and cationic dyes [23]. Fig. 3 shows the general applications of DESs.

8.1. DESs-based analysis of inorganic materials

DESs-based methods have been reported to analyze inorganic



Fig. 3. General applications of DESs.

materials during the present decade. There are different types of analytes such as metals nonmetals and gases. Mao et al. [24] prepared a series of Brnsted-Lewis acidic-based DESs of $C_9H_{10}O_2XZnCl_2$ (X = 0.5 to 3). The extraction agent, catalyst $C_9H_{10}O_2XZnCl_2$ and the oxidant H_2O_2 were used to investigate the desulfurization of a model oil. The effects of the molar ratio of phenylpropanoic acid to zinc chloride, reaction temperature, oxygen to sulfur (O/S) molar ratio, and volume ratio of DESs to oil on the desulfurization rate were studied. A 2:1 phenylpropanoic acid to zinc chloride ratio, a reaction temperature of 50 °C, an O/S ratio of 6, a DESs to oil volume ratio of 0.25:1, and a model oil dosage of 5 mL were found to be the best conditions for desulfurization. Under ideal conditions, a desulfurization rate of 99.23% in 80 min was reached. The DESs' recyclability was also tested, and the results revealed that the desulfurization rate reduced after five runs. These findings show that $C_9H_{10}O_{20}.5ZnCl_2$ DESs have high desulfurization activity and stability in extraction and oxidation desulfurization systems.

8.1.1. Arsenic

Rajput et al. [25] proposed a method for removing arsenite (As^{3+}) species from polluted aqueous solutions utilizing a hydrophobic natural (NADES). THY-DA (1:1), THY-DA (2:1), and THY-DA (1:2) are hydrophobic NADES based on thymol (THY) and decanoic acid (DA). THY-DA (1:2) has better physicochemical qualities than the other two DESs and is thus used as the only extracting agent for removing As^{3+} species in natural water. At ideal extraction conditions (100 mg of DES, 6 min sonication, pH 6.5–7, 25 °C), extraction efficiency (E %) for As^{3+} was greater than 90% for both high (5 mg/L) and low (100 g/L) concentrations. Physicochemical properties of DESs were also examined, including melting point, density, solubility and moisture. The DES was regenerated three times with 0.1 M HCl as a stripping agent, with no significant loss in extraction efficiency (E%). Zounr et al. [26] reported a method for the separation of As(III) and As; a green and highly effective DESs ultrasound-assisted liquid-phase microextraction (DESs-UALPME) method has been used. sodium diethyldithiocarbamate (DDTC) has been used as a chelating agent, choline chloride-phenol as an extracting solvent, and tetrahydrofuran (THF) as a dispersing solvent, As(III) was found to be extracted. Electrothermal atomic absorption spectrometry was used to measure the concentration of arsenic (ETAAS). The developed method was based on large recoveries of As(III) and As(V), and the results were found to be 5%. On the other hand, potassium iodide and ascorbic acid were used to eliminate As(V). This method was useful for figuring out how much arsenic was in the water. While As(V) started as a difference between the total As and As(III) levels. The most important things that affect how well microextraction works are pH, ultrasonication time, and the amount of ligand. Under ideal conditions, the limit of detection (LOD), the limit of quantification (LOQ), the preconcentration factor (PF), and the relative standard deviation (RSD) were calculated to be 10 ng L1, 33 ng L1, 0.25, and 4.3%, respectively. Certified reference materials were used to check the accuracy of the developed technique. As(III) and As(V) in water samples from Tokat, Turkey, were separated by a certain method. It was also used to determine how much total arsenic was in food and the environment. Alomar, M. K., Alsaadi et al., 2016 reported carbon nanotubes (CNTs)-based DES as new materials for extracting arsenic ions (As^{3+}) from water. The benzyltriphenylphosphonium chloride (BTPC) and methyltriphenylphosphonium bromide (MTPB) as salts and glycerol (Gly) as HBD, prepared two DESs systems were made. The prepared material was used for the elimination of arsenic. Different parameters have been used to obtain maximum results. The adsorption efficiency was observed up to 23.4 mg/g.

8.1.2. Chromium

Shi et al. [27] prepared a new family of hydrophobic DESs by mixing parabens and quaternary ammonium. Prepared DESs were used to eliminate Cr (VI) from water samples. It has been observed that hydrophobic DESs display outstanding extraction performance to Cr (VI). When the ratio of DES to Cr (VI) is 1:1 M, the preconcentration factor can reach up to 700, the extraction capacity can reach as high as 66.7 mg g-1, and the extraction efficiency can be better than 90% even for ultra-trace Cr (VI) (0.51 g L-1). The extraction effectiveness remains over 90% even when the concentration of coexisting metal ions (Cd^{2+}, Cu^{2+}) and organic molecules ($(C_6H_{10}O_5)n$, $C_6H_{12}O_6H_2O$) is 2000 times that of Cr (VI). In contrast to the preparation method for ionic liquid extractants, the technique for producing DES is highly straightforward. The rate of atomic utilization is one hundred percent, and no waste is produced. These hydrophobic DESs have been utilized to successfully extract and enrich Cr (VI) from actual water samples. The recoveries achieved were deemed to be satisfactory. FT-IR and 1H NMR are being utilized to study the potential extraction mechanism [28]. Rajput et al. [29]developed a liquid-liquid extraction method for removing Cr(VI) from aqueous samples using a novel green hydrophobic DES plays a role as extracting agent. The hydrophobic DES was created by combining choline chloride and thymol in a 1:4 M ratio and heating it to 70 °C for 10 min (1:4). At optimal reaction conditions (pH 2–6, 40 °C), the ChCl-THY(1:4) eliminate high (20 mg/L) and low (500 g/L) concentrations of Cr(VI) from contaminated water with >95% extraction efficiency. The DES was studied using 1H NMR and FTIR spectroscopy, and the results indicate that interactions occur between the Cl ion of choline chloride and the H atoms of thymol molecules. This study observed no sharp melting point for ChCl-THY(1:4) in the DSC curve. DES was regenerated with 0.1 M NaOH as a stripping agent, and 50–60% extraction efficiency was achieved in the following cycle [30].

8.1.3. Copper

Kanberoglu et al. [31] proposed a method for extracting copper from liver samples using DESs and UA-ELPME. For the digestion and microextraction stages, various types of DESs were prepared. In this procedure, lactic acid and choline chloride was used to prepare DESs, while DESs containing tetrabutylammonium chloride and decanoic acid were employed for ultrasound-assisted emulsification liquid-phase microextraction. Cu(II) ions in the aqueous phase were complexed with sodium dimethyl dithiocarbamate (NaDMDTC). After digestion, the resulting hydrophobic complex was removed to the tetrabutylammonium chloride-decanoic acid based-DES. LOD, LOQ, PF, and % RSD were calculated to be 4.00, 13.2 g L 1, 10%, and 3.2%, respectively. The copper content in liver samples was extracted using the proposed microextraction technique.

8.1.4. Cobalt

Arain et al. [32] reported the method for the pre-concentration of cobalt; a new DES-UALPME technique has been developed. During the preparation of DESs, 1-nitroso-2-naphthol was used as a ligand. Under ideal conditions, the LOD, LOQ, pre-concentration factor (PF), and percent RSD were 1.10 g L 1, 3.60 g L 1, 15, and 7.1%. A micro-sample injection system coupled with a flame atomic absorption spectrometer measured cobalt concentration in DES rich phase (MS-FAAS). The developed method's accuracy was tested by analyzing pharmaceutical supplements and tea samples. Recoveries of certified reference materials and spiked samples ranged from 97 to 105%. Finally, the proposed method's performance characteristics were compared to other liquid phase microextraction procedures.

8.1.5. Lead

Mukhopadhyay et al. [33] reported for the first time that DESs-based adsorbent material was successfully utilized to extract lead from the soil of a landfill. DESs were prepared using choline chloride, a quaternary ammonium salt combined with different HDBs, including sucrose, fructose, glycerol, and ethylene glycol. Saponin derived from the pericarp of soapnut fruits was used to improve the efficacy of DESs. Saponin is a natural biodegradable surfactant called saponin. Lead was removed at approximately 31% and 25%, respectively, by a solution of fruit-based DESs containing fructose and sucrose-containing 10%. The removal rate improved when saponin was added to the solution. A combination of either 10% fructose-based DES and 2% saponin or 40% fructose-based DES and 1% saponin can remove up to 72% of lead. When saponin was added to synthetic DESs, the results showed a significant improvement of up to 54%.

DESs contained both glycerol and ethylene glycol. Mildly alkaline DESs supplied the hydrogen ion, acting as Lewis acid and replacing the lead cations in the organic carbon electron donors. When combined with an acidic saponin solution that provided H+, DESs with a slightly alkaline pH functioned significantly better. This study will open new opportunities for using DESs derived from natural compounds in soil remediation. Mukherjee et al. [34] proposed a DESs-based method for removing lead (Pb) from polluted soil for the first time by utilizing polyol-based DESs combined with a natural surfactant saponin. The DESs in this work were made by combining the quaternary ammonium salt choline chloride with polyols such as glycerol and ethylene glycol. To improve the efficacy of DESs, a natural surfactant saponin extracted from the pericarp of soapnut fruit was combined with them. Because of the higher pH, the DESs did not perform well on their own; nevertheless, they improved the performance of soapnut by up to 100%. Pb removal from contaminated soil was above 72% using a mixture of 40% DES-Gly and 1% saponin and a mixture of 10% DES-Gly and 2% saponin. XRD and SEM examinations revealed no significant deterioration in the soil texture. Because of the environmental friendliness of both DESs and saponin and their low costs, their potential as soil cleaning agents should be thoroughly investigated.

8.1.6. Lithium

Hanada & Goto [35] reported a new environmentally friendly liquid-to-liquid extraction method for lithium (Li) extraction through the synergistic DESs material. The synergistic DESs were prepared using two conventional extractants, a beta-diketone and a neutral extractant. When the two extractants were combined, a liquid DESs was formed. The viscosity of DESs is much lower than that of conventional hydrophobic ionic liquids. The Conductor-like Screening Model for Real Solvents method was used to predict the formation of DESs with melting points lowered due to intermolecular interactions between their components. With the synergistic DESs, the extraction performance of Li was highly improved. Because the optimum DESs combination was solvent-free and highly concentrated, the extraction capacity of Li reached 4.4 g/L. With a model brine solution and the best DESs, the highly selective recovery of Li over Na and K was demonstrated. This first report was highly efficient and selective DES-based liquid-liquid extraction of Li.

8.1.7. Palladium

Tang et al. [36] proposed a method for extracting palladium(II) using Hydrophobic DESs-based material. A new hydrophobic DESs was prepared using the mixture of methyltrioctylammonium chloride ([N8881]Cl) as HBAs and dehydrated fatty acids or fatty alcohols as HBDs. The synthesized DESs were used to eliminate palladium(II) from acidic solutions. Different parameters have been investigated to obtain excellent percentage recoveries, such as DESs species, DESs dose, Contact time, pH study, and temperature. Under the optimal condition, the DESs exhibit a remarkable percentage recovery of palladium(II) in acidic media up to more than 90%. In the meantime, N8881Cl:1-hexanol (1:1) and N8881Cl: hexanoic acid (1:2) display excellent selectivity for palladium(II) over the other coexisting metal ions. In addition, the palladium(II) present in the loaded DES phase was also successfully recovered as elemental palladium using the hydrazine hydrate reduction method. As a result, the DES could be recycled a minimum of five times.

8.1.8. Molybdenum

Zhang et al. [37] developed a new method for eliminating Mo from spent catalysts using hydrophobic DESs. The hydrophobic DESs were synthesized by mixing 1 M of the quaternary ammonium salt with 2 M of saturated fatty acids with varying carbon chain lengths. The extraction ability and mechanism of these DESs were investigated. Different parameters have been optimized during the extraction of Mo, such as the structure of HBAs and HBDs, pH study, agitation time, phase reaction, and temperature, to obtain the maximum percentage recovery. It was observed that the prepared hydrophobic DESs exhibit excellent elimination efficiency toward Mo. The Mo distribution ratio exceeded 2200 in the presence of coexisting metal ions. At the optimal condition, the extraction efficiency was up to 99%. This work reveals a distinct class of materials, guiding an efficient and environmentally friendly method of spent catalyst treatment.

8.1.9. Zinc

Łukomska et al., 2021 [38] proposed a solvent extraction method for extracting zinc and manganese from “black mass” (BM) produced in a zinc plant in Poland, which is secondary waste from spent zinc batteries. Different extraction methods and solvents were investigated, including ILs, organophosphorus-based acids, and DESs. The environmentally friendly method of extracting metal ions without the primary process of leaching waste batteries with sulfuric acid has been proposed. The BM solvent extraction with trihexyltetradecylphosphonium thiocyanate ([P6,6,6,14][SCN]), trihexyltetradecylphosphonium chloride ([P6,6,6,14][Cl], Cyphos IL 101), bis-(2-Ethylhexyl), and bis(2,4,4-trimethylpentyl)phosphinic acid (Cyanex 272), Process variables and organic-to-water ratios were investigated. A flowchart for recovering zinc and manganese from BM has been presented. The extraction efficiency for Zn(II) was 100% with (Cyanex 272 + diethyl phosphite). DES (choline chloride [N_2OH,1,1,1]) was used to obtain a quick and efficient extraction with 100% recovery of zinc and manganese. [Cl]:lactic acid ratio is 1:2, as is choline chloride [N_2OH,1,1,1]. [Cl]: malonic acid (1:1) with the addition of DDACl, a didecyldimethylammonium chloride surfactant. Also, zinc is a metal deposit extracted with organophosphorus-based acids and diethyl phosphite and recovered with potential-control electrolysis from acidic water solutions. The selectivity and distribution ratios for each system have been provided. Zhu et al. [39]reported choline chlorideurea ethylene glycol (ChClureaEG) based DESs to eliminate the zinc from zinc oxide dust. When the slurry concentration is 50 g/L, the leaching temperature is 80 °C, and the stirring speed is 600 r/min, the zinc extraction efficiency can reach 85.2%. Diffusion controls leaching; the related activation energy is 32.1 kJ/mol. The resulting solution was used to electrodeposit zinc directly. A present efficiency of 82.6% is used to acquire the pure zinc deposit. In addition, ChClureaEG DESs can be recycled. This method is effective in recycling zinc from zinc-containing dust.

8.1.10. Collective metals

Ahmadi-Jouibari et al. [40] developed an environmentally

friendly extraction method based on vortex-assisted liquid-phase microextraction, and the solidified DESs (VALPME-SDES) to determine Pb, Cd, Cu, As, and Hg in tea before atomic absorption spectrometry analysis in a graphite furnace (GFAAS). As an extraction solvent, new DESs containing 1-decyl-3-methylimidazolium chloride and n-butanoic acid were utilized, which provided material stability, low density, and a reasonable freezing point near room temperature. The proposed procedure's accuracy was also tested by measuring the concentration of the examined metal ions in a certified Green Tea reference sample (GBW 10052). Spathariotis et al. [41]reported a new method for the selective extraction of iron(III) and zinc(II) from ethaline (1[thin space (1/6-em)]:[thin space (1/6-em)]). 2 M ratio of choline chloride to lactiline (1[thin space (1/6-em)]:[thin space (1/6-em)]ethylene glycol) choline chloride[thin space (1/6-em):[thin space (1/6-em)]lactic acid (2 M ratio), with the former DES being more selective. Iron(III) was selectively removed from a feed comprising zinc(II) and lead using a commercial mixture of trialkylphosphine oxides (Cyanex 923, C923) mixed in an aliphatic diluent (II). Following that, the basic extractant Aliquat 336 was used to separate zinc(II) from lead(II) (A336). For both systems, the equilibration period and extractant concentration were tuned. 1.2 mol L1 oxalic acid and 0.5 mol L1 aqueous ammonia were used to remove iron(III) and zinc(II), respectively. Using commercial extractants, an effective solvometallurgical flowsheet for the separation and recovery of iron(III), lead(II), and zinc(II) from ethaline is presented. The technique was also scaled up in a countercurrent mixer-settler setup, which resulted in successful separation and purification. Ahmadi-Jouibari et al. [40] developed a method for the separation, preconcentration, and detection of cadmium and lead from edible oils, a (DES−LPME) method was developed. An oil sample was treated with a 4:1 mixture of DESs and 2% nitric acid (200 L). The liquid was vortexed before being placed in a 50 °C water bath and agitated for 5 min. After the extraction, the phases were separated by centrifugation, and ETAAS was used to determine the preconcentrated analytes in the DESs phase. Enhancement factors of 195 and 198and limits of detection (defined as 3 Sb/m) of 0. 2 and 8 mg kg1 were reached for cadmium and lead, respectively, under optimum extraction conditions and for a 28 g oil sample. The method successfully determined cadmium and lead levels in different culinary oils. Łukomska et al. [42] describe heavy metal extraction studies from the “black mass” (BM) of waste batteries for the recycling of nickel and cadmium, as well as cobalt (for many samples) from wasted Ni−Cd batteries [43]. DESs and other extractants diluted in toluene or naphtha (kerosene) were used to extract and separate Ni(II) and Cd(II) from solid BM. It has been demonstrated that, in contrast to ionic liquids (ILs), the extraction efficiency of DESs from the BM is 30 wt% Ni(II) and 100 wt% Cd(II) without the need for an initial sulfuric acid leaching process. Solvent extraction has been proposed using two DESs (choline chloride + lactic acid and choline chloride + malonic acid), ionic liquids (ILs) such as trihexyltetradecylphosphonium chloride [P6,6,6,14][Cl] (Cyphos IL 101) and trihexyltetradecylphosphonium thiocyanate [P6,6,6,14][SCN], and organophosphate (D2EHPA). The extraction efficiency and distribution ratio were specified for each system. The metal content of solid BM was determined using SEM/EDS and the UltraWAVE digesting system in conjunction with an atomic absorption spectrometer (FAAS and FAES techniques). The concentration of metal ions in aqueous and stripped organic solutions was determined using an inductively coupled plasma optical emission spectrometer (ICP-OES). Shishov et al. [44] proposed choline chloride and other hydrogen bond donors DESs-based for the elimination of singly- and multiply-charged metals (Al, Ag, B, Ba, Co, Ca, Cr, Fe, Cu, K, Mn, Mg, Mo, Pb, Ni, Sn, Ti, V, and Zn) from oil matrices. Carboxylic acids, polyols, sugars, urea, and their derivatives were investigated as hydrogen bond donors for metal separation in DESs. DESs-based on carboxylic acids and sugars were found to have the maximum extraction efficiency for both singly- and multiply-charged metals extracted from oil matrices. In addition, the effect of water as a DESs precursor on metal separation was examined. The DESs, including choline chloride, lactic acid, and water, had the best extraction recovery of all the solvents tested (up to 99%). Using the ICP-OES method, the extraction solvent was used to analyze oil samples (vegetable oil and motor oil). The detection limits ranged from 0.02 to 17 mg kg1 under ideal conditions. The suggested process does not require microwave digestion of oil samples for elemental analysis. Sorouraddin et al. [45], developed DESs-based highly efficient dispersive liquid-liquid microextraction process for the extraction of Ni and Co. DESs were prepared by mixing the choline chloride (HBD) and 4-aminophenol (HBDs). It's then utilized as both a extraction solvent and chelating agent. The linear ranges for Ni(II) and Co(II) were 0.80−50 and 0.50−50 gL1 under ideal experimental circumstances. The detection limits for Ni(II) and Co(II) were 0.30 and 0.22 g L1.

8.2. DESs-based analysis of organic compounds

Blanco et al. [46] reported DESs based extraction method for removing congo red from water. DESs material has been prepared by combining thymol with menthol, coumarin, octanoic acid, and decanoic acid as extraction agents. The extraction efficiencies of prepared material were higher than 90% for low concentrations of cango red. The best results were obtained when the solvent was a 1:1 mixture of thymol and octanoic acid. Furthermore, each reagent can be reused at the end of the procedure. The reusability of DES was tested for three extraction cycles and found to be stable. Farooq et al. [47] prepared DESs based rose to prominence as a viable alternative to traditional organic solvents and ionic liquids (ILs). They are appealing solvent solutions for extractions and as additives to chromatographic separations because of their adjustable and designer physio-chemical features, low cost, and ease of manufacture. However, DESs contain a broad spectrum of HBA and HBD, so selecting the suitable solvent for separations can be difficult. This article examines all DES preparation methods and explains their benefits and drawbacks. The classification of DESs based on the betaine dye and Nile red scales and Kamlet-Taft characteristics is also covered, as polarity is a key aspect in their employment in separations. Finally, a list of DESs applications in different elimination methods (fuels, phenolics, metals, carbohydrates and proteins), solid-phase extraction, solid-phase microextraction, and capillary electrochromatography is presented [48]. Haider et al. [49] prepared five different hydrophobic natural DESs to extract isopropanol water. Natural hydrophobic DESs prepared by long-lasting solvents with low viscosity (50 mPa s), high thermal degradation temperature (150 °C), and a density differential of more than 50 kg m^3 between water and the DESs. The synthesis of DESs was verified using spectroscopy techniques such as H1NMR and FTIR. The extraction ability of synthesized DESs was then investigated using the pseudo-ternary liquid-liquid equilibrium phase diagram of isopropanol-water-DESs systems. The results reveal that all DESs generate type I ternary LLE diagrams with a wider immiscibility region, positive tie-line slopes, and binodal curves, dependent on the hydrophobic DES precursors utilized. In addition, using the NRTL thermodynamic model, the binary parameters for extracting IPA from water utilizing natural hydrophobic were estimated. Shirazinia et al. [50] prepared various metal complex DESs based on choline chloride (MCDESs) with different metal chlorides such as $FeCl_2$ and $FeCl_3$ $CuCl_2$, $ZnCl_2$ and $SnCl_2$, utilizing 1,3-Diphenylguanidine and 1,5-Diphenylcarbazide as complexing agents and their extractive desulfurization efficiencies

were evaluated. The synthesized MCDESs were characterized using the FT-IR method. Temperature, time, the type of MCDESs, and the mass ratio of MCDESs to model fuel were all evaluated as influential parameters. The best potential for extracting dibenzothiophene, thiophene, and 3-methyl thiophene from n-octane as a model fuel was choline chloride DES based on sn-diphenylcarbazide. After one cycle of extraction, the highest desulfurization (percent) of 95.2 0.3, 94.5 0.5, and 92.1 0.6 were achieved for dibenzothiophene, 3-methyl thiophene, and thiophene, respectively, at optimum conditions of time 10 min, temperature 35 °C, and MCDESs to model fuel mass ratio 1:25, indicating a high capability of the synthesized MCDESs for extractive desulfurization through -complex Furthermore, the used MCDES was effectively regenerated and efficiently reused in the desulfurization process, demonstrating the synthesized MCDESs' industrial potential. Florindo et al. [51] proposed a hydrophobic DESs-based method for eliminating four neonicotinoids—Imidacloprid, Nitenpyram, Acetamiprid, and Thiamethoxam—from diluted water solutions. In particular, two different families of DESs were made, one from natural neutral ingredients (DL-Menthol and natural organic acids) and the other from quaternary ammonium salts and organic acids. Their stability in water was carefully studied using 1H NMR. Only the DESs that are chemically stable were chosen to be used as solvents in the extraction of the four neonicotinoids. This keeps the water cycle clean and allows the DESs to be used again. The final results were compared to those for liquid-liquid extraction using hydrophobic imidazolium-based ionic liquids as solvents. Li et al. [52] reported three different hydrophobic DESs for the extraction of bisphenol DESs were made by mixing tri octyl methyl ammonium chloride, which was used as HBA, with decanoic acid ketoprofen gemfibrozil, which was used as HBD, in different amounts. Prepared hydrophobic DESs were used for vortex-assisted LLME of bisphenols from wastewater. The quantification of bisphenols has been carried out by using the HPLC-FLD. Four bisphenols were chosen as model contaminants: 2,2-bis(4-hydroxyphenyl)butane (BPB), 2,2-bis(4-hydroxydiphenyl)propane (BPA), 4,4′-(1-phenylethylidene)bisphenol (BPAP), and 4,4′-cyclohexylidenebisphenol (BPZ). The enrichment factors of the proposed method were found to be between 97 and 112, depending on how the analytes were made. Under the best conditions for an experiment, the linearity ranges of the method were between 0.3 and 700 g L1, and the linear correlation coefficients (R2) were higher than 0.996. The LODs and LQs were between 0.3 and 0.5 g L1 and 0.06 and 0.08 g L1, respectively. Also, the method was used to eliminate and measure four bisphenols from plastic samples that had come into contact with food. Dogan et al. [53] establish a new, simple, and inexpensive shaker-assisted deep eutectic solvent microextraction (SA-DES-ME) process for extracting paracetamol from synthetic urea and pharmaceutical samples before determining its concentration using a UV−VIS spectrophotometer. The SA-DES-ME approach works by selectively extracting paracetamol using DESs made with choline chloride (ChCI)/betaine (Bet) and various HBDs. The extraction solvent utilized to separate the paracetamol-containing compound from the sample solution was DESs. The univariate optimization method was used to find the best values for the experimental parameters. Under ideal conditions, the suggested approach produced analytical findings such as excellent linearity (50−800 g L1), quantitative recoveries (94.2−107.1%), low detection limit (14.9 g L^1), and a high sensitivity enhancement of pre-concentration factor (135). In addition, the relative standard deviations (RSDs percent) for repeatability and reproducibility analyses were less than 3.3%. By using the standard addition method, the proposed method was successfully used to extract paracetamol from pharmaceutical samples and determine synthetic urea. Phelps et al. [54] developed a method for extracting trace pertechnetate ($99mTcO^-_4$) from an aqueous medium first-time HDESs. A 1:2 M ratio of a tetraalkylammonium or tetra alkyl phosphonium halide combined with a monocarboxylic acid was used to examine the hydrophobic DESs (i.e. saturated fatty acid). Without the use of a formal extracting agent (e.g., ionophore), quantitative (>99%) removal of tracer levels of $99mTc\ O^-_4$ from an aqueous phase in the occurrence of a large excess of competing anions (e.g., Cl^-, NO^{3-}, HCO^{3-}, H_2PO^{4-}, SO^2_4-) in 5−60 min at 25 °C demonstrates an efficient and selective extraction. Pertechnetate distribution ratios above 103 have been seen at infinite dilution. There have also been attempts to remove technetium from the HDES extractant phase by reducing it using $SnCl_2$. Li et al. [9] proposed a new hydrophobic DESs method for removing toluene, ethylbenzene and o-xylene. DESs were prepared by mixing ethylparaben and methyl tri octyl ammonium chloride. Prepared DESs were added to a poly(dimethylsiloxane) (PDMS) fiber sol-gel sorbent coating to make it work better. It was found that adding hydrophobic DESs to the surface of PDMS fibers makes a lot of neat pores, which greatly improved the performance of PDMS fiber coating. For the quantification of volatile organic compounds like toluene, ethylbenzene, and o-xylene pollutants by using GC-FID. The linear range of the PDMS-DES fiber was between 10 and 1000 g L1, and the LOD was between 0.005 and 0.025 g L1. The peak area of signal response for the PDMS-DES fiber was 3 times higher than for the PDMS fiber made at home. At the same time, the LOD for the proposed method was much lower than the LOD for the commercial 100 m-PDMS fiber coating. Gissawong et al. [55] developed a new method for extracting organic pollutants from water and soil samples using DESs melamine sponge (DES−MS). The model pollutants were five organophosphorus pesticides (OPPs): azinphos-methyl (AZP), parathion−methyl (PRT), fenitrothion (FNT), diazinon (DIZ), and chlorpyrifos (CPF), and two dyes: acid blue 29 (AB29) and malachite green (MG). From tetrabutylammonium bromide (TBABr) and different fatty acids, it was easy to make DESs. Before being used to get rid of the studied pollutants, the sponge was loaded with the DESs that had been made. After the removal, high-performance liquid chromatography or derivative spectrophotometry was used to measure the amount of OPPs or dyes left in the supernatant. The proposed method was easy, quick, safe for the environment, and effective, with a removal rate of more than 70% for different samples. Also, the removal of different dyes was done successfully with an efficiency of more than 65% under the best conditions. Blanco et al. [46] developed a method for eliminating Dyes such as thymol, menthol, coumarin, octanoic acid, and decanoic acid that were used to get congo red used newly prepared DESs. Those made by mixing thymol with organic acids had more than 90% extraction efficiencies for low initial concentrations of Congo red when the extraction parameters were 1/5 (v/v) DES/water and 1 h of stirring. Thymol: Octanoic acid (1:1) was the best solvent for getting the best results. Saset al., [56], Developed different DESs-based on thymol, menthol, and organic acid, which are used as separating agents in liquid-liquid extraction to remove 2-nitrophenol and 4-nitrophenol from water. The optimal extraction conditions were studied, and the better extraction yields were obtained using: the (1:1) V: V ratio and 30 min of stirring time. All extraction experiments were carried out at 298.15 K and 30 min of centrifugation time at 6000 rpm. As the results show, these DESs-based on menthol can be used to extract both nitrophenolic compounds, while those based on thymol can only be applied for the removal of 4-nitrophenol. All studied solvents present high extraction efficiencies, over 90%, assuring their capability as extraction agents in aqueous systems. Chen et al. [57] proposed a method to assess the feasibility of capturing VOCs from exhaust using hydrophobic deep eutectic solvents (DESs). Hydrophobic DESs with HBAs and HBDs that are both fatty acids with long alkyl chains (i.e., lauric acid [LauA]) decanoic acid [DecA] and

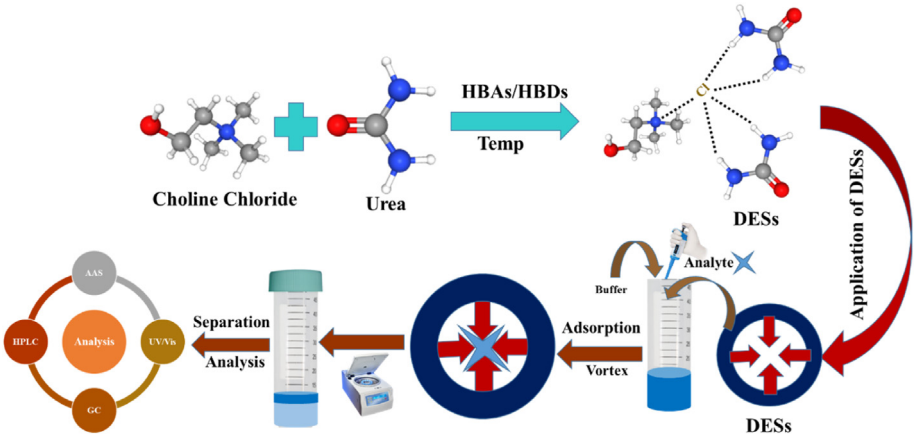


Fig. 4. General adsorption mechanism of DESs.

achieve the highest toluene absorption capacity at 30 °C and 800 ppm toluene with an HBA: HBD molar ratio of 1:3 Because toluene solubility increases with the alkyl chain length of the fatty acid eutectics, hydrophobic interactions between the fatty acids and toluene are critical for improving the VOC absorption ability of DESs. The compiled results indicate that the proposed hydrophobic fatty acid eutectics, [LauA][DecA], can achieve effective hydrophobic VOC removal, and the physical absorption nature of toluene in fatty acid eutectics benefits easy recovery and recycling. Fig. 4 represents the general adsorption mechanism of DESs.

9. Conclusion

Adsorption is essential in analytical procedures, mainly when the environmental sample analysis or sample matrices are complex. Analytical performance is significantly impacted by selecting the best adsorbent solvent. The DESs have a broad spectrum of applications in the separation sciences, not just for their "greenness" but also for their extraordinary physicochemical properties. The DESs-based adsorbent has excellent molecular recognition efficiency in the aqueous phase. It is prepared in the aqueous media, escaping the consumption of a large quantity of traditional organic solvents. Due to its outstanding property, it simultaneously plays a porogen and imprinting path for the targeted analyte. Because of its ionizable nature,in the DESs the stronger hydrogen bonding, offers unique behaviour from the traditional solvents when used in the extraction process. However, the DESs-based adsorbents applications are still deficient, and the advantages of DESs over the outdated solvents have not been fully revealed. So that more efforts should be made to examine better DESs species for chromatographic separation techniques. It will be stimulating to explore the DESs adsorbents-based new environmental applications.

10. Future aspects

The DESs-based adsorbent materials need more attention and research to explore the new modifications and applications in the separation sciences. DESs-based adsorbent materials have been used successfully to extract and separate a wide range of analytes like aromatic hydrocarbons, biodiesel, gases, metal, and metal. Still, researchers focused on modifying mobile and stationary phases for chromatographic techniques such as GC and HPLC; additionally, DESs should be used for the sample preparation for chromatographic determination. So it can be predicted that the use of DESs should be enhanced to advance new robust and chromatographic methods. In this prospect, it must be explored the temperature effect and the effect of water quantity on the DESs preparation and stability.

It is important to emphasize different applied fields such as electrochemistry, separation sciences, pharmaceuticals and drug delivery if we expect that DESs will be used in analytical chemistry in ways other than as extractants. The use of DESs in this field has enormous potential that is essentially unexplored. Only a little bit of work has been completed in this area. Additionally, more research needs to be done on how DESs affect chromatography separation, including the potential use of DESs as a stationary phase in gas chromatography, a capillary modifier, or a mobile phase in electrophoresis as substitutes for ionic liquids. A wide range of work is also made possible by synthesizing and modifying adsorbents in DESs media for additional applications in analytical chemistry. The scope of study in this field virtually expands to infinity when we consider that it is possible to combine several areas, such as the fabrication of MIPs on the surface of electrodes in a DES medium. Therefore, we suggest that analytical chemistry researchers concentrate on using the potential application of DESs in the separation sciences and employing them in various areas. In actuality, DESs have not yet been used in conjunction with such directions as microfluidic analysis, the creation of paper microchips, luminescence and chemiluminescence, surface plasmon resonance, electrophoresis, sampling, and sample preparation. In the future, combining nanotechnology with DESs needs deep exploration and applications in separation sciences. Combining nanotechnology and DESs will open new doors in the separation sciences.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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